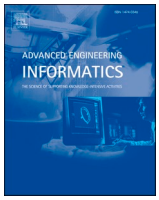
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Hybrid monitoring methodology: A model-data integrated digital twin framework for structural health monitoring and full-field virtual sensing

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ABSTRACT

This study presents a structural full-field sensing framework consisting of multi-type and multi-scale models. The digital twin is established by integrating finite element models (FEMs), bridge information model (BRIM), and internet of things (IoT). Multi-scale FEMs are built as the mechanical twin model to recognize the hidden features and information from the monitored data, where the hybrid monitoring methodology is adopted to realize the conversion, expansion, and fusion of data. Meanwhile, the BRIM serves as the data twin model to collect the life-cycle information from designing to operation, where a data visualization interface is developed to present heterogeneous information. Moreover, the physical entity and cyberspace are connected and interact via the IoT system. A paradigm is finally developed for intelligent big-data management and heterogeneous data fusion in structural health monitoring. It is applied to a three-span continuous model bridge for verification.

1. Introduction

Bridge structures worldwide require intensive maintenance throughout their service life, because they are facing increasing hazards caused by environmental loads and climate changes. "*Canadian Infrastructure Report Card in 2019*" suggested that almost 40 % of the civil assets investigated are in fair or worse condition, merely 20 % of which were constructed in the last two decades [1]. "*American Infrastructure Report Card 2021*" highlighted that 7.5 % (46154) of bridges were designated as defective, and the cost of annual rehabilitation increased to $22.7 billion to improve their serviceability [2]. The long-term serviceability of bridges depends not only on the quality of construction but also on the level of management and maintenance [3]. In this regard, intelligent information techniques can improve the health status of bridges and address new challenges in bridge management.

Defect detection and operational state assessment are important tasks in bridge management. Conventionally, visual inspection can detect surface defects to evaluate the states of bridges [4], whereas, it is subjective, inefficient, and inaccurate [5]. In recent years, structural health monitoring (SHM) has developed as a promising tool for intelligent management [6] through the full life cycle of bridges, including the design, construction, management, and maintenance stages [7]. SHM is advanced by building information modeling (BIM), numerical modeling, internet of things (IoT), big data, and automatic technologies [8]. In particular, BIM is a revolutionary technology that significantly improves the performance and applicability of an SHM system from the following two aspects [9].

First, BIM provides an information integration and communication platform for bridge SHM. BIM includes rich semantic information in a common data environment to digitalize industry products [10]. Specifically, the cloud-based BIM was developed to facilitate a high level of cooperation among project participants with various knowledge and backgrounds [11]. It provides a digital collaboration space for information gathering, managing, and sharing throughout the lifecycle of engineering projects. In this regard, BIM serves as a repository to store massive data associated with SHM, such as those about objects, documents, inputs, outputs, IoT, management tools, etc. [12,13].

Second, BIM serves as a data visualization platform. The customized visualization interface can be developed for information visualization, structure assessment, and data management based on the secondary development function of BIM software. In this process, 4D visualization is essential to represent the long-term changing states of bridges, where 4D indicates that 3D information at different time moments is dynamically analyzed and presented with the additional time dimension. For

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example, Zanen et al. proposed an impact assessment method for the construction of infrastructures, where the spatial data is integrated with the construction plans and progresses to achieve 4D visualization [14]. Zhou et al. proposed a 4D model to manage heterogeneous information about the building components and construction progress, where the in-site vision was used to generate an augmented reality model to facilitate communication between workers and managers [15]. Zhang et al. integrated BIM and construction schedules to achieve 4D management and safety control [16].

The IoT system can enhance and support a BIM because they are complementary. BIM represents physical entities with a high spatial resolution, while IoT collects real-time information with a high temporal resolution [17]. Some studies highlighted the improved efficiency by integrating BIM and IoT systems, which enabled real-time feedback for further plans [18,19]. The BIM-IoT system provides a novel way for data collecting, storing, and exchanging. It consists of three parts: 1) contextual information can be stored in BIM tools or IFC formats, including the building geometry, IoT system, and static and soft information; 2) the recorded time-series data from sensors are queried by structured query language; 3) the integration between contextual information and time-series data should be realized. The most critical aspect of the BIM-IoT system lies within the third part, and five integration methods have been concluded in Ref. [17]. However, these contents are rarely applied to a real bridge [20–22].

A serious limitation of the BIM-IoT system is that BIM does not reflect the mechanical property of infrastructures, which is the essence of SHM. Besides, massive data collected via IoT should be processed and analyzed to mine valuable information. The hybrid monitoring methodology developed in the past five years can overcome these limitations. Hybrid indicates the integration of data and model, where a finite element model (FEM) is used to analyze the measured data to find hidden features and identify potential risks. In this paper, the hybrid monitoring-related works are summarized, and a data-FEM-BIM-IoT digital twin model is developed for bridge SHM. The in-field bridge, FEM, and BIM are connected via IoT in a unified framework, and the overall connections of each component are validated via a set of experimental tests. Moreover, data and information flow between a real bridge and virtual models in dual directions, and thus, the physical and cyber spaces interact in a real-time manner. The highlights of hybrid monitoring include:

- A data-model integrated paradigm for SHM is developed and applied to a model bridge, to realize the digitalization of infrastructures.
- The spatially discrete data monitored is virtually expanded to the structural full domain, improving the density, diversity, and integrality of information.
- Heterogeneous information and multi-structure data are fused to improve the performance of current SHM systems.

The rest of the paper is arranged as follows. The literature about digital twins is reviewed in Section 2, followed by a digital twin architecture developed in Section 3. Subsequently, the hybrid monitoring methodology is briefed in Section 4. The hybrid monitoring-based digital twin is applied to an in-field bridge model in Section 5. The conclusions are outlined in Section 6.

2. Literature review on digital twins

The concept of digital twin emerged from the aerospace projects in 1970, when a high-fidelity simulator assisted in developing an optimal plan in a rescue mission [23]. The definition of digital twin was later given by Michael Grieves in 2003 as “equivalent digital representations of physical entities” [24]. However, it was not widely noticed until 2010 when NASA decided to build digital twins for future vehicles [25,26]. From then on, digital twin started to become mature with an increasing application value. From 2017 to 2019, digital twin was consecutively listed as the top ten strategic technology trends. Meanwhile, it is regarded as a critical method to realize Industry 4.0 because it involves virtual simulation and information fusion. In civil engineering, digital twin was also applied to the full lifecycle of buildings, including construction, operation, repair, maintenance, and demolition.

Digital twins enable efficient planning, implementation, and management in civil construction projects. For example, Kaewunruen et al. [27] simulated the construction process of London King Cross station by using a digital twin, converting a 3D model to a 6D information model that includes geometry, time, cost, and carbon emission. Pan et al. [11] proposed a digital twin framework including BIM, IoT, and data mining to achieve intelligent construction management. Zhang et al. [28] applied digital twins to field management to refine the detailed processes, which was based on BIM, IoT, and digital data representation. Moreover, digital twins can help urban-level management from operation to maintenance via risk-predicting and decision-making. Ham et al. [29] created a risk-sensing-based urban digital twin model to evaluate the urban vulnerability to extreme weather events. The digital twin system was also used in various aspects of smart cities, such as those in transportation [30,31], energy systems [32], public safety [33], and healthcare [34].

The modeling methods have been investigated and advanced to achieve high-fidelity digital twins in recent years. For example, Lu et al. [35] applied a slice-based object fitting method to a reinforced concrete highway bridge, which accurately converted the collected point clouds to a digital model. Jiang et al. [36] modeled a digital twin of an old highway from map data, where the horizontal and vertical curves, cross-section shapes, vehicle-lane arrangement, and central protection zones of the road were well considered and saved. Jia et al. [37] proposed a novel digital twin strategy with standardized model division and assembly processes, which could be applied to complex structures with multi-scale geometry, multi-type information, and various application scenarios.

Moreover, digital twins are enhanced via intelligent technology such as machine learning [38]. Austin et al. [39] combined the digital twin with a semantic model-based deep learning architecture to establish an intelligent platform for urban sensing and decision-making. Chakraborty et al. [40] investigated a physics and data co-driven framework for multi-timescale dynamic digital modeling. The challenges in realizing digital twins were summarized by Broo et al. [41] from the aspects of system, information, and organization, emphasizing the multidisciplinary cooperation to develop intelligent structures.

Despite the applications of digital twins in engineering projects, they are rarely applied to bridge SHM cases [42]. Digital twin technique is widely adopted in the design [43–45] and construction [46–48] of a bridge, but limited in operation and maintenance [49]. In the literature, the digital twin technology is used for bridge inspection using unmanned aerial vehicles, [3], dynamic tests, [50], and component coding techniques [51]. However, current digital twin frameworks lack a real-time data analysis and processing modulus for the long service life of bridges [52]. Moreover, they cannot jointly model the geometry information, load-response characteristics, and mechanical behavior of a bridge. To fill these research gaps, this paper has developed an integrated bridge digital twin framework. The hybrid monitoring methodology serves as an efficient data analysis modulus to improve the performance of the framework. An in-field bridge model subjected to ambient and vehicular loads is adopted for verification.

3. A digital twin framework

The digital twin framework is introduced in this section, as displayed in Fig. 1. It consists of three components, the physical entity, FEM, and bridge information model (BRIM). Dependent physical and cyber entities are connected via IoT and communication systems. The physical entity is the monitored bridge, the response of which to ambient and traffic loads is measured by distributed sensors. The measured data are

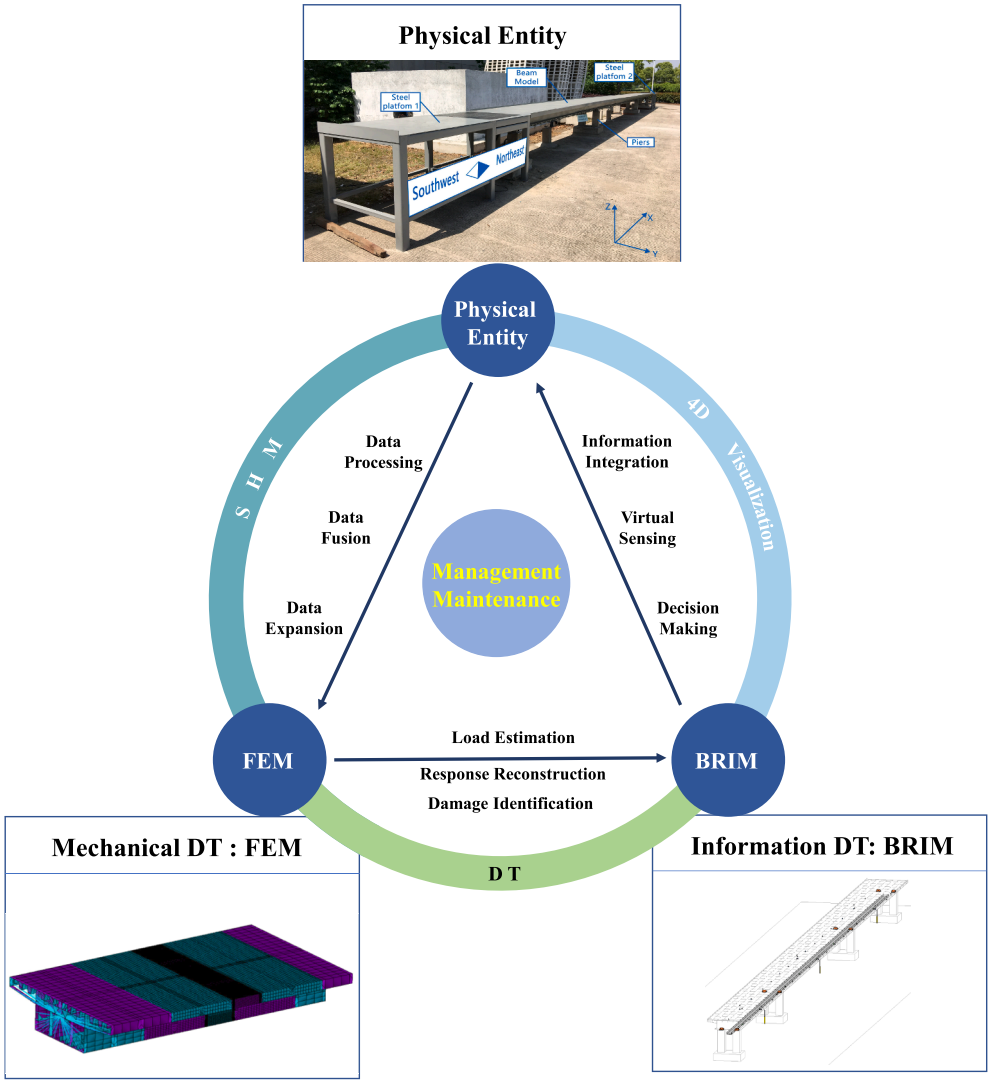


Fig. 1. Digital twin framework for bridge SHM.

then collected and analyzed in a service center. Subsequently, the FEM is adopted to realize the fusion, separation, conversion, and expansion of data. After that, the essential information is saved and exhibited in a BRIM, and necessary actions are fed back to the real bridge. In this process, three components interact with each other to form a closed loop. The developed framework is based on the FEM and information model, which are available for other types of structures. Therefore, it is applicable and scalable for other mechanical systems, such as buildings, dams, wind turbines, etc. The bridge model is here used as a representative example.

The physical entity is the base of a digital twin, namely, the cyberspace is initially created from such a prototype. On the other hand, the in-service bridge is also the end of the developed framework because its state is unknown and needs to be assessed after being changed by factors such as degeneration, maintenance, and repair. To be specific, a bridge is subjected to ambient, traffic, and corrosion effects, and thus, its health state degenerates at a slow rate over a long period. Except for the natural aging process, the bridge can also be suddenly damaged after extreme events, including earthquakes, typhoons, over-weighted vehicles, etc. These continuous and sudden degenerating effects are investigated in SHM by measuring the loads and structural responses via IoT. Whereas, the measured data is spatially discrete and insufficient given the huge volume of a bridge. Besides, IoT could be affected by magnetic environments, making the measured data contaminated. The noised data are collected, preprocessed, and further analyzed to grasp the health state of a bridge.

FEM is used to simulate and analyze the mechanical behavior of the real bridge. Its element stiffnesses are updated according to the monitored data to identify the potential damage or risks. Moreover, FEM is used to calculate the full-field bridge responses to different loads, which is an essential step to realize the holographic virtual sensing of a real bridge. In this process, the FEM and load distribution should both be accurate. The major difficulty lies in monitoring the load because it includes many different components, such as the wind, traffic, pedestrian, and temperature loads, whose spatial distributions are complex and unknown. The hybrid monitoring theory has provided a set of load identification and approximation methods to estimate the temporal and spatial distributions of loads. The structural responses to the real and estimated loads are close. In this way, the full-field responses are calculated to achieve virtual response sensing, and the monitored space-discrete data is expanded to the full bridge [53].

BRIM is a data integration modulus fusing multi-source and multi-structure data, including the sensor attribute, structural response, load distribution, damage condition, and periodical inspection text or pictures. Heterogeneous data are first analyzed by data-driven approaches. Subsequently, they can be visualized and managed in a unified framework re-developed in BIM software. The temporal evolution and spatial features of a bridge can be analyzed in the BRIM modulus in a 4D

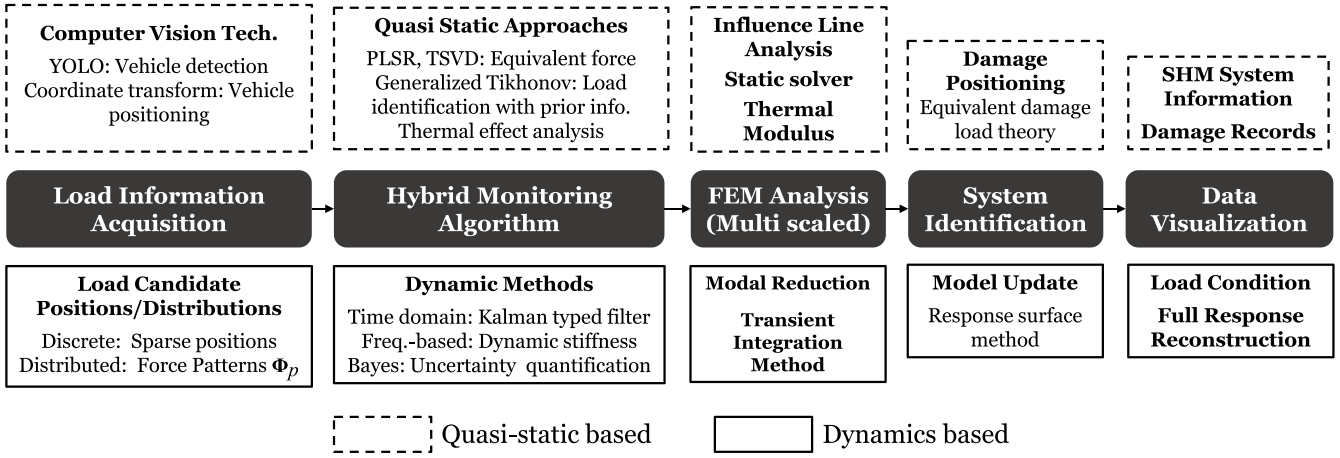


Fig. 2. Summary of hybrid monitoring.

manner given the data accumulated over a long period, where the full-field 3D data from FEM calculation are visualized dynamically with the additional time dimension aided by the hybrid monitoring approaches. Moreover, the integration of heterogeneous data can help to make reasonable management decisions, for example, the optimal chance of the closure or repair of a bridge.

IoT is used to connect the physical entity, FEM, and BRIM. The sensor network measures the synchronized load and response data of the in-service bridge. These data are then transmitted to the central servicer to realize the fusion, separation, conversion, and integration of information by the hybrid monitoring approaches. Subsequently, the integrated information is exhibited to the users in the BRIM, and necessary actions are consecutively applied to the real bridge, such as the temporary closure of a bridge, reduction of the traffic density, mechanical component repair, etc. when the responses are abnormal or over-limit in cyberspace. Then, the state of the real bridge would be changed, and correspondingly, the FEM and BRIM should be updated by using the information from the real world during the mentioned feedback actions. Based on IoT and information transmission systems, the cyber and physical models interact actively and bidirectionally, realizing the digitalization of bridges.

4. Hybrid monitoring

Hybrid monitoring is a model-data-co-driven methodology to advance bridge SHM. It considers the structural load distribution, real-time responses, damage condition, and safety state. As the bridges are subjected to abundant types of load, the involved data is heterogeneous. Hybrid monitoring is divided into three parts: static deformation sensing, dynamic response monitoring, and damage identification. The used techniques and developed approaches are summarized in Fig. 2.

4.1. Multi-data fusion-based quasi-static response monitoring

For simplicity, hybrid monitoring started from studying the quasi-static structural responses to different types of long-period loads, where the dynamic effect is minor and negligible. Heterogeneous data are involved and fused in this section to study different types of loading effects. Besides, the thermal effect is considered because it is a long-period load.

4.1.1. Quasi-static load estimation

The structural deformation to quasi-static loads can be calculated by using the influence line approach. When a load $\mathbf{F} \in \mathbb{R}^{n\times 1}$ (n is the number of DOFs of the FEM) is applied to the bridge, the static deformations are represented as

$$\begin{aligned} \boldsymbol{\varepsilon} &= \mathbf{EF} \\ \mathbf{d} &= \mathbf{DF} \Rightarrow \mathbf{Y} = \mathbf{WF} \\ \mathbf{r} &= \mathbf{RF} \end{aligned} \tag{1}$$

where $\varepsilon \in \mathbb{R}^{N_\varepsilon}$, $\mathbf{d} \in \mathbb{R}^{N_d}$, and $\mathbf{r} \in \mathbb{R}^{N_r}$ are strain, displacement, and rotation observation vectors, respectively; $\mathbf{E} \in \mathbb{R}^{N_\varepsilon \times n}$, $\mathbf{D} \in \mathbb{R}^{N_d \times n}$, and $\mathbf{R} \in \mathbb{R}^{N_r \times n}$ are corresponding influence line matrices; and N_ε, N_d, and N_r denote the numbers of sensors. $\mathbf{Y} \in \mathbb{R}^{N}$ represents the observation vector. $\mathbf{W} \in \mathbb{R}^{N\times n}$ is the input-to-output matrix. The load is assumed unknown or partially known because the load monitoring technique is still immature and limited in bridge SHM.

As the measured data is spatial-discrete and deficient, load estimation is ill-posed where a small perturbation in the measurement may lead to a significant difference in the estimated loads. The concept of equivalent load is developed to estimate the unknown load vector, and two methods were adopted, the partial least square regression (PLSR) [54] and truncated singular decomposition (TSVD) [55]. PLSR and TSVD can both address ill-posedness in load estimation. PLSR is introduced first.

The static governing equation of a mechanical system can be formulated as follows when the observation is noisy:

$$\mathbf{Y} = \mathbf{WF} + \mathbf{e} \tag{2}$$

where $\mathbf{e}$ is the noise vector. Eq. (2) has the same form to a regression equation:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e} \tag{3}$$

where $\mathbf{Y}$ is the dependent vector, $\mathbf{X}$ is the independent matrix, and $\boldsymbol{\beta}$ is the regressional coefficient vector. PLSR can estimate $\boldsymbol{\beta}$ efficiently from noise data when the number of the columns of $\mathbf{X}$ is far greater than that of the rows. Therefore, PLSR can also estimate the load vector in Eq. (2).

The governing equation (1) can also be solved by TSVD. The input-to-output matrix $\mathbf{W} \in \mathbb{R}^{N\times n}$ can be decomposed as follows using orthogonal matrices $\mathbf{U} \in \mathbb{R}^{m\times m}$ and $\mathbf{V} \in \mathbb{R}^{n\times n}$:

$$\mathbf{W} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^{\mathrm{T}} \tag{4}$$

where

$$\boldsymbol{\Sigma} = \begin{pmatrix} diag(\sigma_1, \sigma_2, \ldots, \sigma_r) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$$

where σ_r is the rth singular value of $\mathbf{W}$, and $\sigma_1 \geq \sigma_2 \geq \sigma_3, \cdots, \sigma_r$. The unimportant singular values after σ_i are truncated and set as zeros. The load vector $\mathbf{F}$ is then calculated by using the pseudo-inverse operator as follows:

$$\mathbf{F} = \mathbf{V}\boldsymbol{\Sigma}^{+}\mathbf{U}^{\mathrm{T}}\mathbf{Y} \tag{5}$$

After the load vector is estimated by using PLSR or TSVD, the full-field quasi-static responses can be reconstructed by applying the loads to an FEM.

4.1.2. Multi-structure data fusion

An SHM system includes heterogeneous data measured by various sensing systems. These data have different structures and formats, and they should be fused and integrated to obtain reliable results. The response data, such as deflection, rotation, strain, and internal forces, can be directly formulated in the governing equation (1) because they are time series with an identical format. Whereas, heterogeneous data is involved in SHM. For example, traffic information is widely monitored by using video systems, which has different data structure from the measured response data. The video data can be used to position the vehicles on the bridge deck. Such information about load should also be considered in the load estimation framework to improve the performance of hybrid monitoring. Two approaches are developed by formulating the load prior information as the generalized regularizations.

Above all, the open-source YOLO network for object detection is used to identify the vehicles on the bridge deck, where YOLO is the abbreviation of "you look only once". A pixel-to-physical transformation function is adopted to calculate the vehicular positions. Then, the load prior information vector is formulated as

$$\mathbf{F}_{prior} = [\mathbf{0}_{U-\Omega}, \mathbf{1}_{\Omega}] \tag{6}$$

where U is a full set including all DOFs and Ω is a subset containing DOFs where the vehicles are located. The unknown magnitudes of nodal forces in Ω are assumed as unit one in Eq. (6). Then, the measured structural response data is used to identify the load vector. A generalized Tikhonov regularization method [56] and a sparse Bayesian approach [57] were proposed.

The generalized Tikhonov regularization approach is briefed first. The object function is formulated as follows given the load prior information vector $\mathbf{F}_{prior}$:

$$\mathbf{F} = \arg\min\left(\|\mathbf{W}\mathbf{F} - \mathbf{Y}\|_2^2 + \lambda\left\|\mathbf{F} - \mathbf{F}_{prior}\right\|_p^p\right) \tag{7}$$

where λ is a regularization coefficient determining the trade-off between the residual term and the regularization term. $p = 2$ is selected due to the convex nature and mathematical conciseness of the quadratic function. Then, the objective function is solved by setting its partial derivative with respect to $\mathbf{F}$ to be zero. The optimal load vector is represented as

$$\mathbf{F} = \left(\mathbf{W}^{\mathrm{T}}\mathbf{W} + \lambda\mathbf{I}\right)^{-1}\left(\mathbf{W}^{\mathrm{T}}\mathbf{Y} + \lambda\mathbf{F}_{prior}\right) \tag{8}$$

The calculated load vector $\mathbf{F}$ is set as $\mathbf{F}_{prior}$ in the next iteration, and the load vector can be iteratively estimated ultimately. In this process, the regularization coefficient λ is a key hyperparameter determining the performance of the approach. It can be selected by using the L-curve method or cross-validation criterion.

Moreover, a Bayesian methodology is developed to select reasonable and optimal λ automatically when the noise magnitude is unknown. The posterior probability function of the load $\mathbf{F}$ given noisy observation $\mathbf{Y}$ is formulated as follows by considering the probability functions of the load and measurement:

$$p(\mathbf{F}|\mathbf{Y}) = \frac{p(\mathbf{Y}|\mathbf{F})p(\mathbf{F})}{p(\mathbf{Y})} = cp(\mathbf{Y}|\mathbf{F})p(\mathbf{F}) \propto p(\mathbf{Y}|\mathbf{F})p(\mathbf{F}) \tag{9}$$

where $p(\mathbf{Y}|\mathbf{F})$ is the likelihood function, $p(\mathbf{F})$ is the prior probability function, and $p(\mathbf{Y})$ is a normalization constant c^{-1}. The load vector $\mathbf{F}$ can be estimated via maximizing a posterior (MAP) probability as follows:

$$\mathbf{F} = \left(\mathbf{W}^{\mathrm{T}}\mathbf{W} + \alpha\boldsymbol{\Omega}^{-1}\right)^{-1}\mathbf{W}^{\mathrm{T}}\mathbf{Y} \tag{10}$$

where α is the variance of noise and $\boldsymbol{\Omega}$ is the covariance matrix of the load vector, and they are iteratively estimated by the MAP strategy. In the Bayesian case, the noise variance parameter α is equal to the regularization coefficient λ in Eq. (8).

4.1.3. Thermal effect

Bridges are subjected to thermal load throughout their service life. The thermal effect could be significant especially when the temperature tremendously changes. Thermal load is also studied in hybrid monitoring [58]. The temperature variation of the bridge girder can be decomposed into uniform and gradient ones. The uniform temperature variation causes longitudinal deformation while the gradient one leads to bending, and they both result in strain responses.

The longitudinal strain response at the sensor positions due to the uniform temperature variation is expressed as

$$\boldsymbol{\varepsilon}_U = \mathbf{1}\cdot P_U \tag{11}$$

where P_U is a constant relating to the material property and shape of the cross-section. For simplicity, merely the vertical temperature gradient is considered. The distribution function of the gradient temperature is denoted as T_G, the strain response ε_G to which is represented as

$$\varepsilon_G(x,z) = \left(G(x)\int T_G(z)zdz\right)z = P_G \tag{12}$$

where x and z are longitudinal and vertical coordinates, respectively. G is a coefficient depending on the material property, cross-section, and boundary condition. Without loss of generality, the temperature gradient T_G is assumed consistent along the longitudinal direction, and T_G-induced strain responses at the sensor positions are expressed as

$$\boldsymbol{\varepsilon}_G = P_G(\boldsymbol{\varepsilon}_B)_{n_\varepsilon \times 1} \tag{13}$$

where $\boldsymbol{\varepsilon}_B$ is the strain observation vector when the baseline temperature field is applied. $\boldsymbol{\varepsilon}_B$ is measured or calculated by an FEM.

After the uniform and gradient temperatures are considered, the load-to-strain mapping equation becomes

$$\boldsymbol{\varepsilon}_{n_\varepsilon \times 1} = P_U\mathbf{1} + P_G\boldsymbol{\varepsilon}_B + \mathbf{E}\mathbf{F}_{n\times 1} \tag{14}$$

Two coefficients P_U and P_G relating to the temperature field should be estimated by the PLSR and TSVD methods. In this way, the thermal load is regarded as an additional load type in hybrid monitoring.

4.2. Dynamic response hybrid monitoring

The traffic, seismic, and wind loads all excite bridge vibrations. The dynamic responses are also studied in hybrid monitoring. Two approaches are investigated: the dynamic stiffness-based offline (frequency-domain) method and the filter-based online (time-domain) method. Both methods are based on the dynamic equilibrium equation:

$$\mathbf{M}\ddot{\mathbf{x}}(t) + \mathbf{D}\dot{\mathbf{x}}(t) + \mathbf{K}\mathbf{x}(t) = \mathbf{F}(t) \tag{15}$$

where $\mathbf{M}$, $\mathbf{D}$, and $\mathbf{K} \in \mathbb{R}^{n\times n}$ are mass, damping, and stiffness matrices, respectively; $\mathbf{x} \in \mathbb{R}^n$ represents the displacement vector; and t represents discretized time. Eq. (15) is reformulated in other forms to study the bridge dynamics.

4.2.1. Dynamic stiffness-based off-line approach

Dynamic stiffness is a frequency-based method. When the structural response time history is measured, the Fourier transformation is used to calculate the frequency responses. The dynamic equilibrium equation is then represented as

$$\left(-\omega^2\mathbf{M}+i\omega\mathbf{C}+\mathbf{K}\right)\mathbf{X}\{\omega\}=\mathbf{F}\{\omega\} \tag{16}$$

where $(-\omega^2\mathbf{M}+i\omega\mathbf{C}+\mathbf{K})\in\mathbb{C}^{n\times n}$ is the dynamic stiffness matrix. The form of Eq. (16) at an arbitrary discrete frequency ω is identical to that of Eq. (1) in the static case, such that the load $\mathbf{F}\{\omega\}$ can be estimated using the static load estimation approaches, which are PLSR or TSVD.

The ill-posed condition of load estimation would be alleviated by reducing the dimension of the load vector $\mathbf{F}$. A set of load distribution vectors $\mathbf{\Phi}_p\in\mathbb{R}^{n\times p}$, derived from the impulse response matrix in a time step, is introduced to approximate the randomly distributed loads [59]. The governing equation (16) is then approximated as

$$\left(-\omega^2\mathbf{M}+i\omega\mathbf{C}+\mathbf{K}\right)\mathbf{X}\{\omega\}=\mathbf{\Phi}_p\mathbf{S}\{\omega\} \tag{17}$$

where $\mathbf{S}\{\omega\}\in\mathbb{C}^{\mathrm{p}}$ denotes the magnitudes of the load distribution vectors of $\mathbf{\Phi}_p$ at a discrete frequency ω. As p is less than the dimension of the observation vector N, $\mathbf{S}\{\omega\}$ can be estimated by using the least square operator. After that, $\mathbf{F}\{\omega\}=\mathbf{\Phi}_p\,\mathbf{S}\{\omega\}$. In this way, the frequency load vector $\mathbf{F}\{\omega\}\in\mathbb{C}^{\mathrm{n}}$ is equivalently represented by p load distribution vectors of $\mathbf{\Phi}_p$. Besides, the estimated load is less affected by observation noise because $p<N$, and the ill-posed condition is consequently improved.

Based on the dynamic stiffness method, an error propagation approach is developed to quantify the uncertainty of the reconstructed responses due to observation noise. Without loss of generality, the observation noise is assumed as the white noise. Its power spectrum is equal to $2\sigma^2\Delta t$ at an arbitrary discrete frequency, where Δt is the sampling interval and σ^2 is the variance of noise. The observation noise propagates to the reconstructed responses via the estimated load. The uncertainty of the reconstructed displacements is calculated as

$$\mathbf{X}_e=2\sigma^2\Delta t\left(\mathbf{H}\mathbf{\Phi}_p\right)\left[\left(\mathbf{H}\mathbf{\Phi}_p\right)^{\mathrm{T}}\left(\mathbf{H}\mathbf{\Phi}_p\right)\right]^{-1}\left(\mathbf{H}\mathbf{\Phi}_p\right)^{\mathrm{T}} \tag{18}$$

where $\mathbf{H}$ is the frequency response matrix.

The frequency-domain methods are concise and straightforward in theory. Whereas, the calculating efficiency reduces significantly when a long period of data is considered, such that the time-domain approaches should be adopted.

4.2.2. Filter-based online approach

The dynamic equilibrium equation is reformulated as follows in a time-discrete state-space form:

$$\mathbf{u}(t_{k+1})=\mathbf{A}\mathbf{u}(t_k)+\mathbf{B}\mathbf{F}(t_k) \tag{19}$$

where $\mathbf{u}=[\mathbf{x},\dot{\mathbf{x}}]^{\mathrm{T}}\in\mathbb{R}^{2n}$ is the state vector, $\mathbf{A}\in\mathbb{R}^{2n\times 2n}$ is state transition matrix, and $\mathbf{B}\in\mathbb{R}^{2n\times n}$ is the impulse response matrix in a time step.

The time-invariant linear system model and observation model are formulated as

$$\begin{aligned}\mathbf{u}_{k+1}&=\mathbf{A}\mathbf{u}_k+\mathbf{B}\mathbf{F}_k+\mathbf{v}_k\\ \mathbf{Y}_k&=\mathbf{G}\mathbf{u}_k+\mathbf{J}\mathbf{F}_k+\mathbf{w}_k\end{aligned} \tag{20}$$

where $\mathbf{G}\in\mathbb{R}^{N\times 2n}$ is the state-to-observation (indirect feedthrough) matrix and $\mathbf{J}\in\mathbb{R}^{N\times p}$ is the input-to-observation (direct feedthrough) matrix. $\mathbf{v}_k\in\mathbb{R}^{2n}$ represents the system noise that is an independent Gaussian process with zero mean and covariance of $\mathbf{Q}\in\mathbb{R}^{2n\times 2n}$. $\mathbf{w}_k$ is the observation noise vector with zero mean and independent covariance of $\mathbf{R}\in\mathbb{R}^{N\times N}$.

The dimension of the load vector $\mathbf{F}$ is still far greater than that of the observation vector $\mathbf{Y}$. $\mathbf{F}$ should thus be compressed such that it can be estimated by using the Kalman-type filters, for which the load distribution vectors $\mathbf{\Omega}_p$ can still be adopted. As an alternative, the candidate load locations can also be assigned to several spatially discrete positions out of all DOFs.

The Kalman-type filters were widely studied in response reconstruction, including the augmented Kalman filter and Dual Kalman filters. By contrast, the joint input-state estimator (JISE) developed by Gilligns et al. is a promising tool in simultaneous load-response reconstruction, and it is verified to outperform conventional Kalman-type filters. Therefore, JISE is adopted in hybrid monitoring [60]. JISE includes three steps: input estimation, measurement update, and time update. The procedures of JISE are summarized in Algorithm 1. JISE requires known covariance matrices $\mathbf{Q}$ and $\mathbf{R}$, and the invertibility of matrices in Algorithm 1 should be ensured. After addressing these issues, both the load and response time histories are accurately reconstructed.

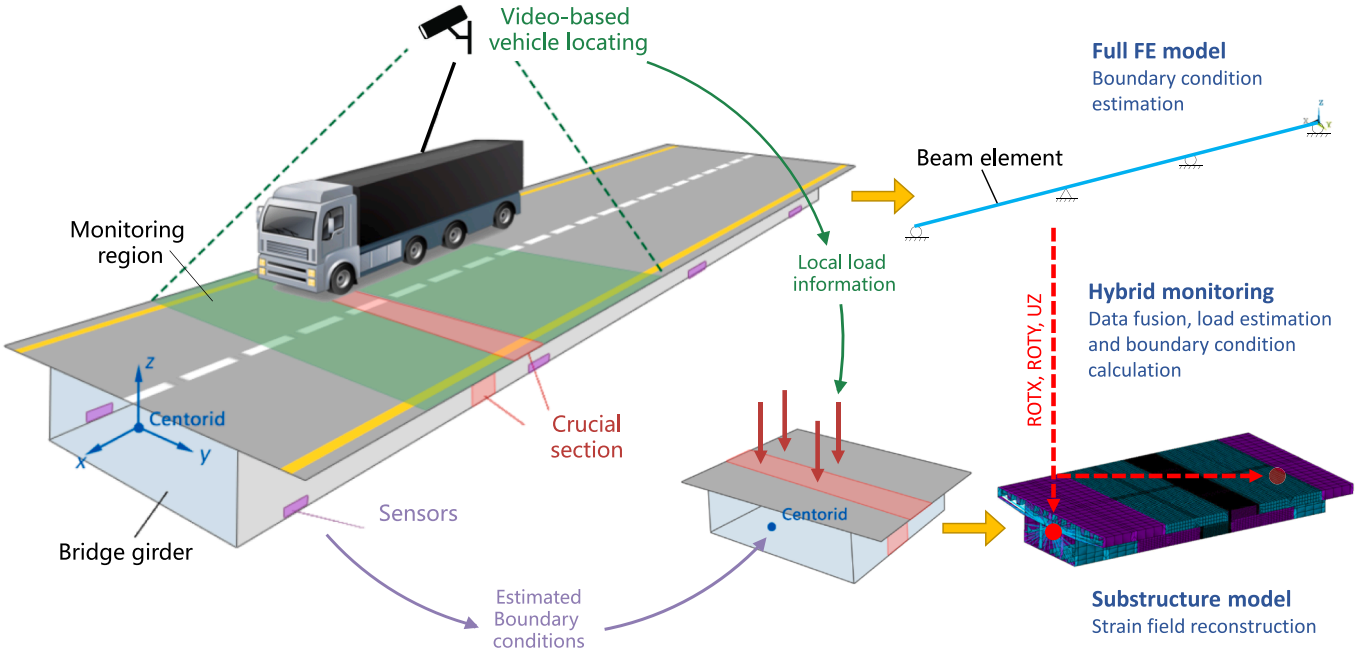


Fig. 3. Schematic of hybrid computing.

Algorithm 1. (*Procedures of JISE*)

(1) Input estimation.

$$\begin{aligned}\widetilde{\mathbf{R}}_k &= \mathbf{G}\mathbf{P}_{k|k-1}\mathbf{G}^{\mathrm{T}} + \mathbf{R}\\ \mathbf{P}_{\mathbf{p}|k|k} &= \left(\mathbf{J}^{\mathrm{T}}\widetilde{\mathbf{R}}_k^{-1}\mathbf{J}\right)^{-1}\\ \mathbf{M}_k &= \mathbf{P}_{\mathbf{p}|k|k}\mathbf{J}^{\mathrm{T}}\widetilde{\mathbf{R}}_k^{-1}\\ \mathbf{p}^*_{k|k} &= \mathbf{M}_k(\mathbf{d}_k - \mathbf{G}\mathbf{u}_{k|k-1})\end{aligned} \quad ,(21)$$

(2) Measurement update.

$$\begin{aligned}\mathbf{L}_k &= \mathbf{P}_{k|k-1}\mathbf{G}^{\mathrm{T}}\widetilde{\mathbf{R}}_k^{-1}\\ \mathbf{u}_{k|k} &= \mathbf{u}_{k|k-1} + \mathbf{L}_k(\mathbf{d}_k - \mathbf{G}\mathbf{u}_{k|k-1} - \mathbf{J}\mathbf{p}^*_{k|k})\\ \mathbf{P}_{k|k} &= \mathbf{P}_{k|k-1} - \mathbf{L}_k(\widetilde{\mathbf{R}}_k - \mathbf{J}\mathbf{P}_{\mathbf{p}|k|k}\mathbf{J}^{\mathrm{T}})\mathbf{L}_k^{\mathrm{T}}\\ \mathbf{P}_{\mathbf{xp}|k|k} &= \mathbf{P}^{\mathrm{T}}_{\mathbf{px}|k|k} = -\mathbf{L}_k\mathbf{J}\mathbf{P}_{\mathbf{p}|k|k}\end{aligned} \quad ,(22)$$

(3) Time update.

$$\begin{aligned}\mathbf{u}_{k+1|k} &= \mathbf{A}\mathbf{u}_{k|k} + \mathbf{B}^*\mathbf{p}^*_{k|k}\\ \mathbf{P}_{k+1|k} &= [\mathbf{A}_k\ \mathbf{B}_k]\begin{bmatrix}\mathbf{P}_{k|k} & \mathbf{P}_{\mathbf{xp}|k|k}\\ \mathbf{P}_{\mathbf{px}|k|k} & \mathbf{P}_{\mathbf{p}|k|k}\end{bmatrix}\begin{bmatrix}\mathbf{A}_k^{\mathrm{T}}\\ \mathbf{B}_k^{\mathrm{T}}\end{bmatrix} + \mathbf{Q}\end{aligned} \quad .(23)$$

4.2.3. Multi-data fusion and hybrid computing

After collecting heterogeneous information about the load and structural responses, they are fused in a hybrid computing framework consisting of multi-scale FEMs to reconstruct the local strain field [61]. The schematic is shown in Fig. 3. A rough-meshed model is used to calculate the boundary condition of the substructure, meanwhile, a fine-meshed substructure model is adopted to calculate the detailed strain responses in a crucial or fragile region.

The beam or link elements are used to build a global FEM, based on which an equivalent load is calculated from the measured global responses. Subsequently, the global displacement time history is reconstructed by using FEM analysis, and the time-varying boundary condition of the substructure is obtained. Then, shell or solid elements are used to build a refined substructure FEM to realize the accurate simulation. The monitored or estimated loads are applied to this substructure model to reconstruct the local strain field via FEM analysis. Moreover, the substructure model can be further refined to analyze the stress concentration effect.

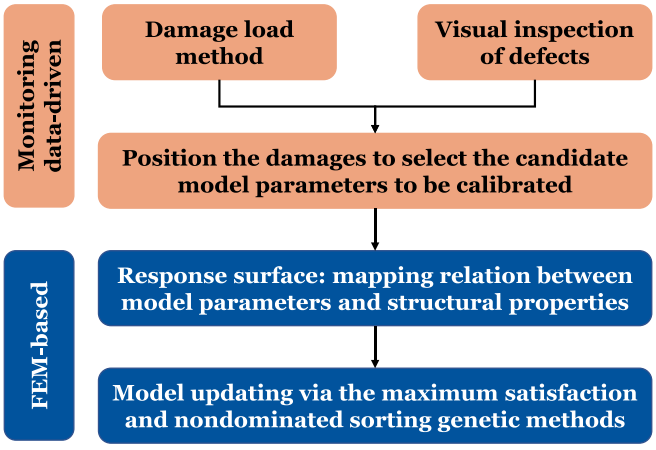


Fig. 4. Procedures for model updating.

4.3. Damage sensing and model updating

The operation state of bridges is changing in their service life. The bridges may be permanently damaged subjected to earthquake, typhoon, and over-limit vehicular loads. Therefore, identifying damages timely and updating the FEM correctly are necessary to keep the digital model consistent with its physical counterpart.

The damage-locating method is introduced first. Based on the developed load estimation methods, a damage load approach is proposed to identify the damaged area. The damage load is derived by using a uniform cross-section beam with elastic supports [62]. The definition of the damage load is simple and direct: the deformation of the undamaged beam to the damage load equals the deformation difference between the damaged and undamaged beams to an arbitrary-distributed load. Analytical results indicate that the damage load is zero and nonzero in the intact and damaged areas, respectively. It is thus a good damage indicator to locate the damage.

After the damaged region is identified, a response surface method is developed to update the FEM [63], as shown in Fig. 4. First, an objective function is selected to calibrate the candidate model parameters. It consists of the natural frequency, mode shapes, and deflection influence lines. Subsequently, the candidate parameters are combined in an ergodic manner to calculate the structural properties in the objective function, which forms a high-dimensional response surface (or mapping relationship) between the candidate parameters and the structural properties. Finally, the candidate model parameters are calibrated by minimizing the objective function using the maximum satisfaction method and nondominated sorting genetic algorithm. In this way, the FEM is updated continuously to reflect the health condition of a real bridge.

5. A case study of the hybrid digital twin

The developed techniques and approaches were applied to a three-span bridge model to establish a digital twin. The bridge model, IoT system, FE model, and information model are introduced in this section. They are connected within the developed hybrid monitoring framework.

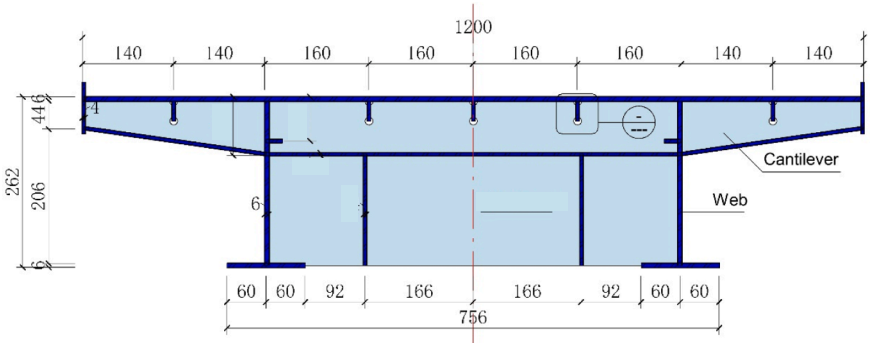


Fig. 5. Cross-section of the main girder.



Fig. 6. Photo of the bridge model.

5.1. Introductions of the bridge model and IoT system

The bridge model was made of 6 mm thick steel plates. The material was Q235 steel in Chinese code. The elastic modulus was 205 GPa and the density was 7850 kg/m^3. The span configuration was 6 + 10 + 6 = 22 m. The girder cross-section is π-shape with the width and height of 1200 mm and 262 mm, respectively, as shown in Fig. 5. The transverse beams or diaphragms were welded to the main girder every 0.4 m in the longitudinal direction to increase its torsional stiffness. The model was placed outdoors and subjected to different types of environmental loads, as shown in Fig. 6.

Four recovery damages were preset, as shown in Fig. 7. The lower flanges of the girder at the damage positions were connected using steel plates and friction bolts. The connecting plates were removed and the lower flanges became broken to simulate the damages. The damages located at $x = -7.1, -2.1, 2.1$, and 7.1 m ($x = 0$ m at the bridge centre), and the corresponding damage lengths were 200, 100, 50, and 20 mm, as denoted in Fig. 7b.

The bridge model was equipped with a monitoring system consisting of sensors, a transmission system, and a data acquisition instrument. A total of 74 sensors were installed, including three displacement transducers, 11 inclinometers, 32 strain gauges, 11 accelerometers, eight reaction force meters, and nine thermometers. These sensors were installed at 21 longitudinal positions, as summarized in Table 1. The sensor positions on the cross-section are illustrated in Fig. 8. Most of the sensors, except for accelerometer A11, were used to measure the structural response in the longitudinal-vertical plane. Besides, a weather

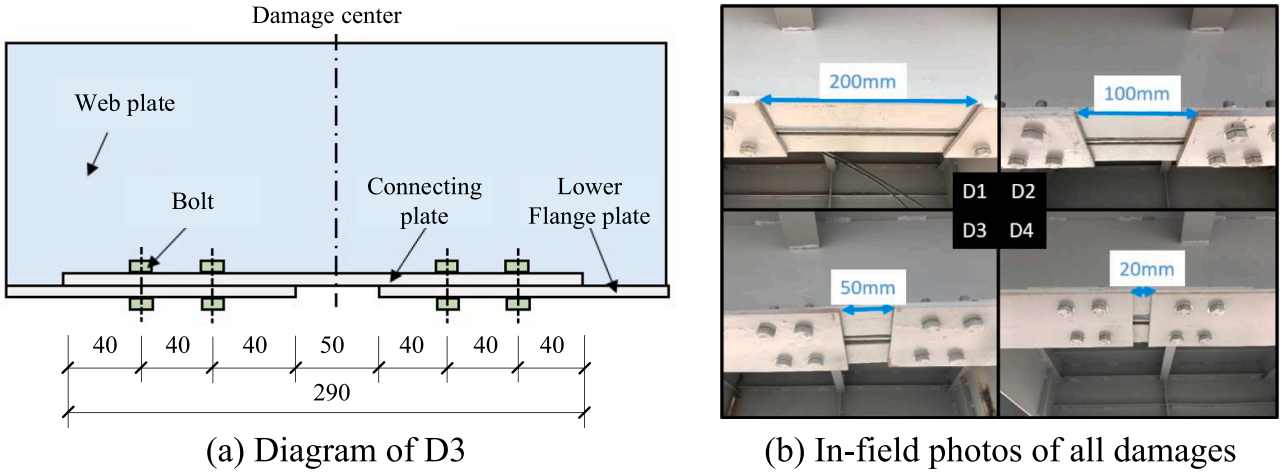


Fig. 7. Preset recovery damages.

Table 1
The configuration of sensors (D and R denote the displacement transducer and inclinometers, respectively).

Section No.	x	Sensor type	D	R	Strain gauge				Accelerometer		
		Position	a	b	c	d	e	f	g	h	i
1	− 10.4	Sensor No.		R01							
2	− 9.6				S01	S02					
3	− 8		D01	R02	S03	S04	S27	S28	A01	A02	
4	− 6.4				S05	S06					
5	− 5.6			R03							
6	− 4.8				S07	S08					
7	− 4.4			R04							
8	− 3.6				S09	S10					
9	− 2.4			R05					A03	A04	
10	− 1.2				S11	S12					
11	0		D02	R06	S13	S14	S29	S30	A05	A06	A11
12	1.2				S15	S16					
13	2.4			R07					A07	A08	
14	3.6				S17	S18					
15	4.4			R08							
16	4.8				S19	S20					
17	5.6			R09							
18	6.4				S21	S22					
19	8		D03	R10	S23	S24	S31	S32	A09	A10	
20	9.6				S25	S26					
21	10.4			R11							
Total No.	3	11	32	11							

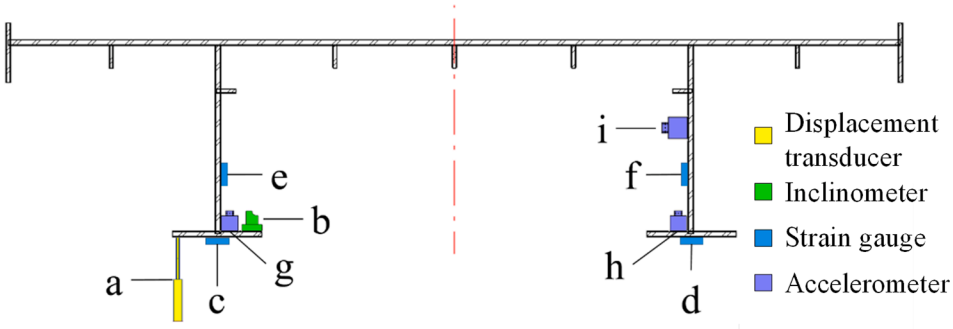


Fig. 8. Placement of sensors in a cross-section.

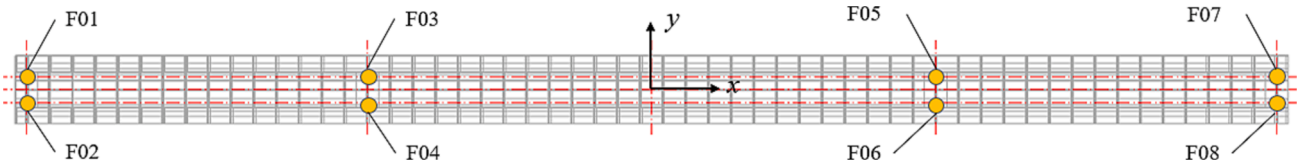


Fig. 9. Placement of reaction force meters.

station, a weighbridge, and a video camera were installed. All these sensors have composed the IoT system for sensing the real-time structural responses to complex loads (See Fig. 9).

5.2. Multi-degree and multi-scale FEMs

FEM is the kernel in the hybrid monitoring framework to realize load estimation, full-field sensing, and damage identification. Four FEMs were built and calibrated to satisfy different demands [64], as shown in Fig. 10: Model 1 consists of 110 beam elements, which is efficient in calculation; Model 2 is a shell model with high accuracy but low calculating efficiency; Model 3 is a multi-scale model balancing the accuracy and efficiency, where shell elements are used between − 2.7 and 2.7 m [65–67]; Model 4 is a substructure model for local strain field reconstruction, and the meshing size near the focused section is refined.

The FEMs are adopted to accomplish different goals of hybrid monitoring. The beam model 1 has minimum DOFs. It is used to estimate the longitudinal load distribution and reconstruct the displacement and

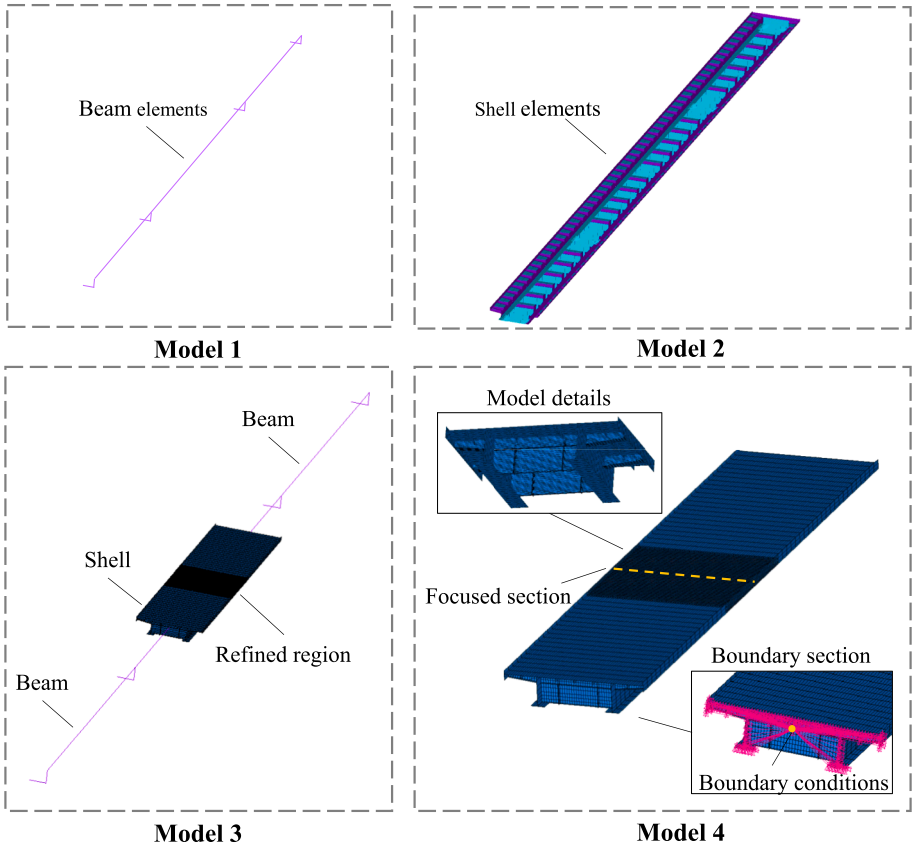


Fig. 10. FEMs used to establish the mechanical digital twin.

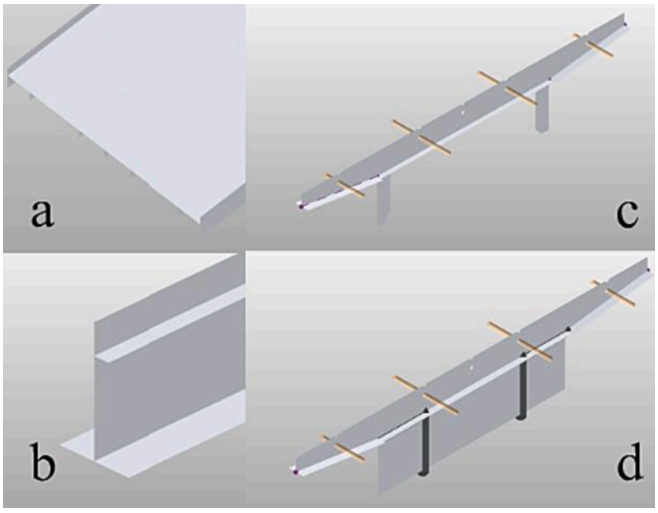


Fig. 11. Model details in SketchUp.

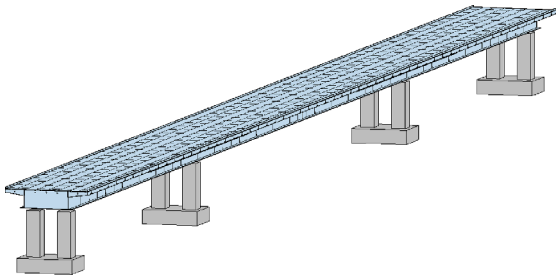


Fig. 12. Overview of the geometric model in BRIM.

acceleration responses. Besides, the potential damage is identified by using this model because its ill-posed condition is the best among all models. Model 2 is a shell FEM to accurately reconstruct the full responses. Models 3 and 4 are used to reconstruct the strain field in the crucial region. Their difference lies in that when using Model 4 the boundary condition of the substructure should be calculated in advance, while the load distribution out of the substructure is not needed. If the defects were identified, all models should be calibrated to keep consistent with the real bridge.

5.3. Bridge information model

Bridge SHM is associated with heterogeneous information, such as structural mechanics, IoT data, load and response information, damage condition, operation state, etc. A BRIM is adopted to integrate heterogeneous data of the studied model bridge. The Revit software is used to build the BRIM with the secondary development function.

Above all, the geometric model should be built in the Revit software by assembling the bridge components of a customized family. The detailed components of the bridge girder were first established in the SketchUp software, as shown in Fig. 11, including the top plate (a), web plate (b), transverse beam (c), diaphragm (d), and stiffeners, etc. These components were subsequently imported to Revit as a volume family with assigned material properties. The 3D geometric model of the bridge was subsequently built by assembling the customized components with a sufficient level of detail, as shown in Fig. 12.

The IoT system was then virtually digitalized in cyberspace. Different types of sensors were modeled in the customized sensor families. A sensor family includes two sets of parameters, namely, the type parameter and instance parameter, as denoted in Fig. 13. The shared information in a sensor family is defined as the type parameter, while the unique or specific information of each sensor is defined as the instant

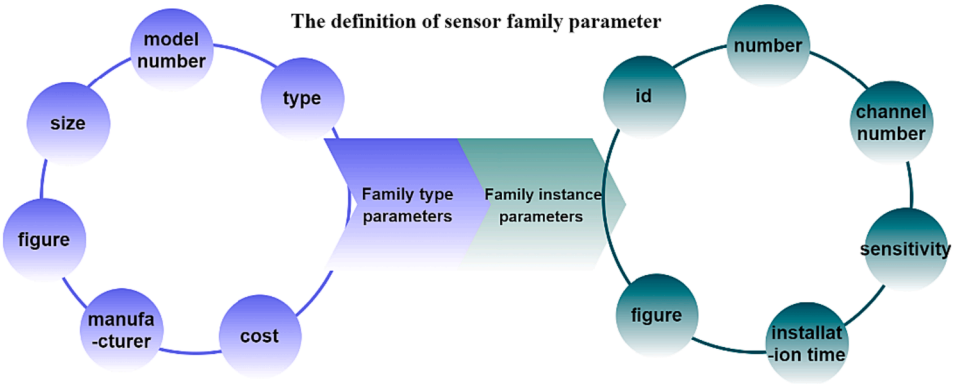


Fig. 13. Classification of sensor parameters.

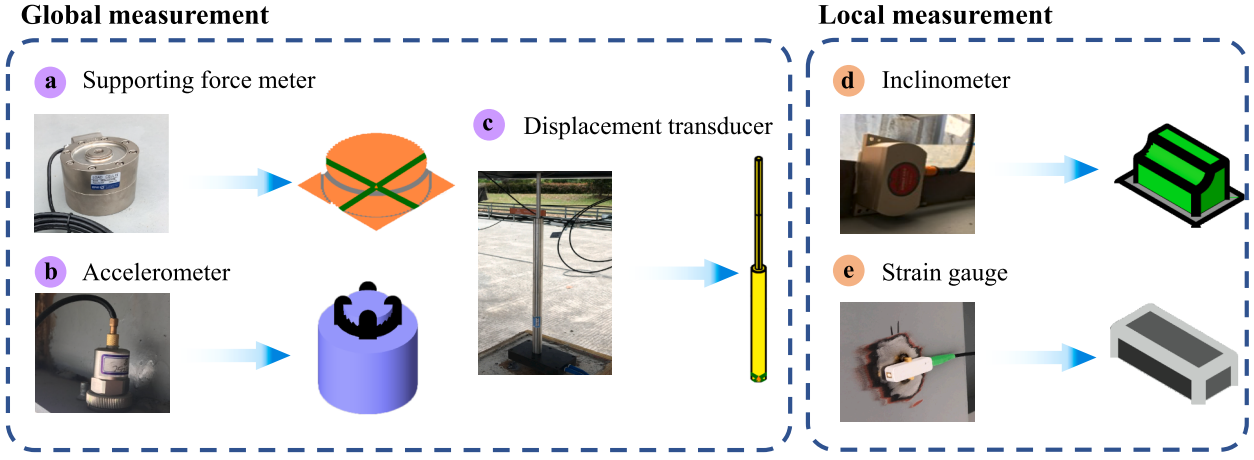


Fig. 14. Creation of the cyber models of sensors.

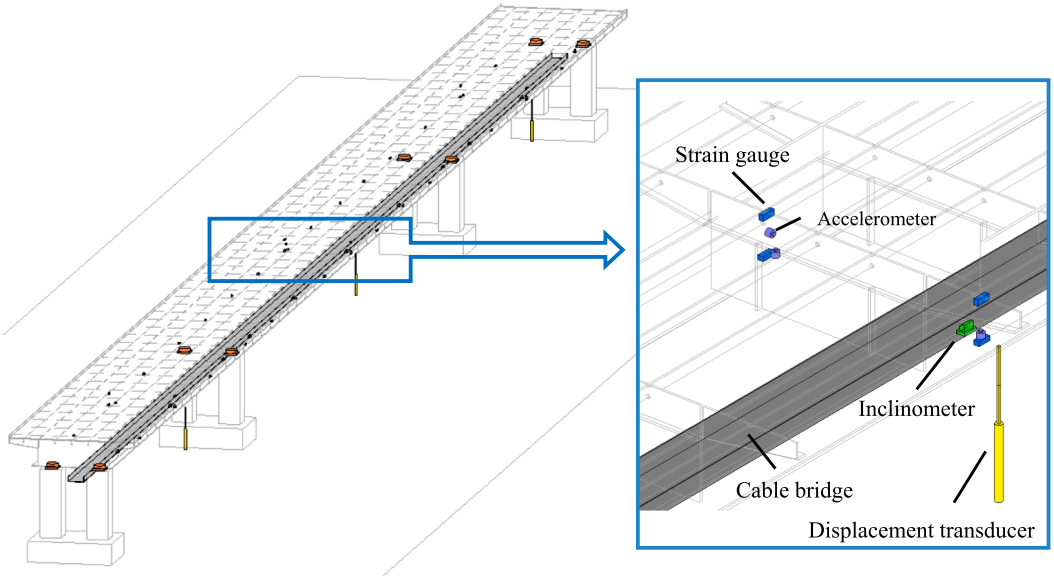


Fig. 15. Integrated BRIM.

parameters. The type parameters of the sensor family include the sensor type, sensor style, manufacturer, cost, etc. The instance parameters include the sensor ID, sensor positions, channel ID, sampling frequency, frequency response range, sensitivity, etc.

The real sensors were digitalized in Revit as instances, as illustrated in Fig. 14. The cyber sensors were then added to the digital geometric model as another family. The type and instance parameters were assigned to each sensor. In this way, massive information associated with the IoT system is stored and managed in an easy-access file. The bridge model and installed sensors were integrated and visualized in cyberspace to improve the data management efficiency, as illustrated in Fig. 15.

5.4. Data fusion and visualization

After the BRIM is established in cyberspace, the monitored heterogeneous data, including the video, environment, identified load, and response data, are fused and visualized on the interface of Revit based on its secondary development function [68,69]. In this process, different software is used to analyze and manage the associated data. The numerical simulation software Matlab is used to program the hybrid monitoring algorithms, the mechanical analysis software Ansys is used to conduct FEM analysis, and the general programming software Python is used to automate the data transmission among Matlab, Ansys, and Revit.

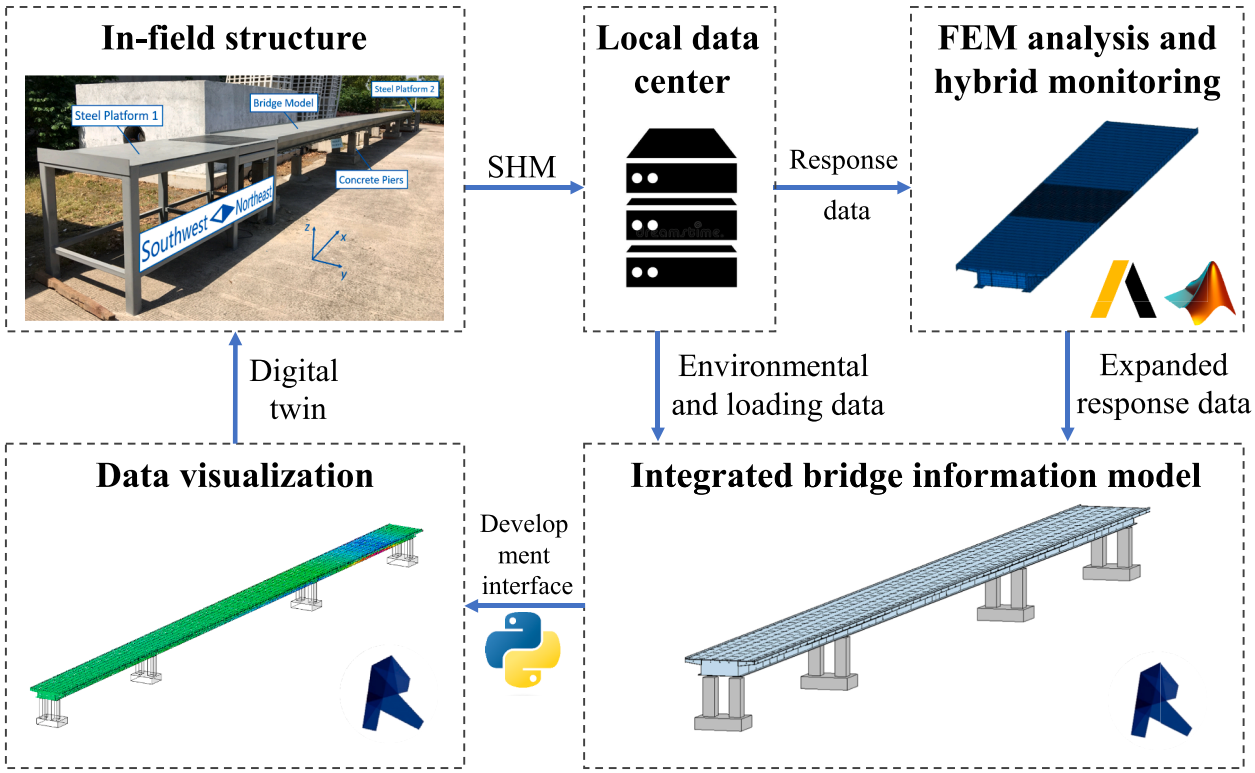


Fig. 16. Workflow of data visualization based on hybrid monitoring.

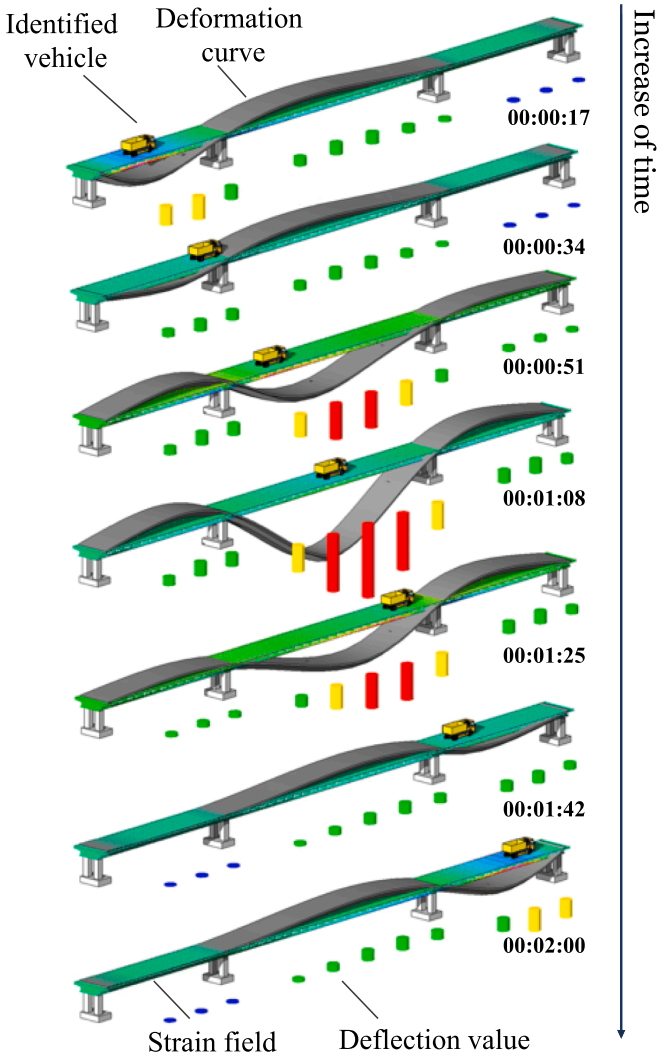


Fig. 17. Dynamic 4D visualization of structural response.

The workflow of data integration is illustrated in Fig. 16. The model bridge was equipped with an SHM system. The measured data are first collected in a local service center, where the video data is used to position the vehicles (wheels) on the bridge deck based on computer vision techniques. Subsequently, the measured structural responses are fed into the hybrid monitoring framework to estimate the load distributions and reconstruct the full-field responses. Combining the identified vehicle locations and measured (or estimated) axle weights, the local strain response field can be calculated accurately using a refined shell FEM. After the FEM analysis, the reconstructed strain and displacement data are extracted and saved in an independent file. Then, they are transformed into a data cloud format and mapped to the geometric model in BRIM. Finally, the strain field, vertical deflection curve, and traffic load are integrated into one interface to achieve multi-data fusion and visualization, as shown in Fig. 17. The root mean square error of the reconstructed responses is below 5 % compared with the ground truth in most cases [54]–[64]. In the quasi-static vehicular load condition of Fig. 17, the averaged root mean square error of the identified load is 3.80 %, that of the reconstructed deflection response is 1.73 %, and the maximum error of strain response is 10.01 $\mu\varepsilon$ at the center cross-section. In the dynamic pedestrian-running load condition, the maximum deflection error is 0.39 mm [60]. In the thermal load condition lasting for 5 days, the maximum error of the thermal response is 0.30 mm [58]. Only the results of the quasi-static loading condition are displayed here for brevity.

Heterogeneous data are transferred between each component in Fig. 16. The monitored data is stored in text format and accessed by the numerical computing software. The structural load and responses are calculated and stored in text format again. Then, all data in text format are read by the data visualization platform. Switching between different software is realized by Python Language without privacy and human intervention (such as controlling the opening and closing of different software, and adjusting the input and output on different language platforms). As the data is stored every 1 min, this framework has a 1-min latency. The response time for load estimation (Matlab), FEM calculation (Ansys), and data visualization (Revit) are 0.002 s, 0.2673 s, and 1.527 s, respectively, for each time step with the computer configuration of Intel Core i7-7700HQ processor, NVIDIA GTX 1060 graphics card, and 8 GB DDR4 RAM.

The bridge structures could be damaged under environmental and traffic loads. In the developed digital twin framework, a new damage family is used to manage the damage information accumulated in a long service time. The damage information is saved in the BRIM when a defect is identified. It would also be updated if the damage condition changed. Each damage instance includes individual instance parameters, such as the size, detection time, photo, and type of damage, as shown in Fig. 18. After a long period, the records of damages can be retrieved and analyzed for bridge management. In such a process, manual efforts are needed to quantify the damage extent and detail the damage information. The automation level of damage modeling is insufficient in this framework and needs improvement in the future.

6. Concluding remarks

The hybrid monitoring-based digital twin is developed in this study to virtually sense the structural state. The FEM, BRIM, and IoT are integrated into a uniform framework for efficient data management and 4D visualization. A novel paradigm of data fusion and information integration is established for SHM.

The hybrid monitoring methodology combines the monitored data and FEM analysis to dig out hidden information and features. Multi-source response data are fused to estimate the load distribution and reconstruct the full displacement field. The traffic and environmental data are incorporated with the substructure technique for local strain field reconstruction. Heterogeneous data can be expanded, converted, and integrated to detect potential damages for model updating. The life circle data, from design to operation, are involved and considered within the developed BRIM.

The proposed digital twin framework was successfully applied to a large-scale bridge model. The bridge structure and its IoT system are completely digitalized in a BRIM to manage massive heterogeneous data. Aided by hybrid monitoring, a multi-scale FEM group is established as the mechanical digital twins, reflecting the structural state from different aspects and in various scales. The FE analysis results and in-field monitoring data are integrated and visualized in an interface that is secondarily developed in the BRIM, realizing the multi-source, multi-structure, and multi-dimension data fusion. The developed framework and connections among each component have been validated via a set of experiments.

In addition to its merits, this study has several limitations. The whole framework uses multiple software and abundant time is needed in the interaction and transmission of data among software. Besides, an updated and reliable finite element model is the basis of the framework, which is not always available in practice. In such a case, more data-driven approaches should be resorted and integrated. Moreover, the real bridges are much more complex than the model bridge. The framework should be applied to an in-service bridge for validation in the future.

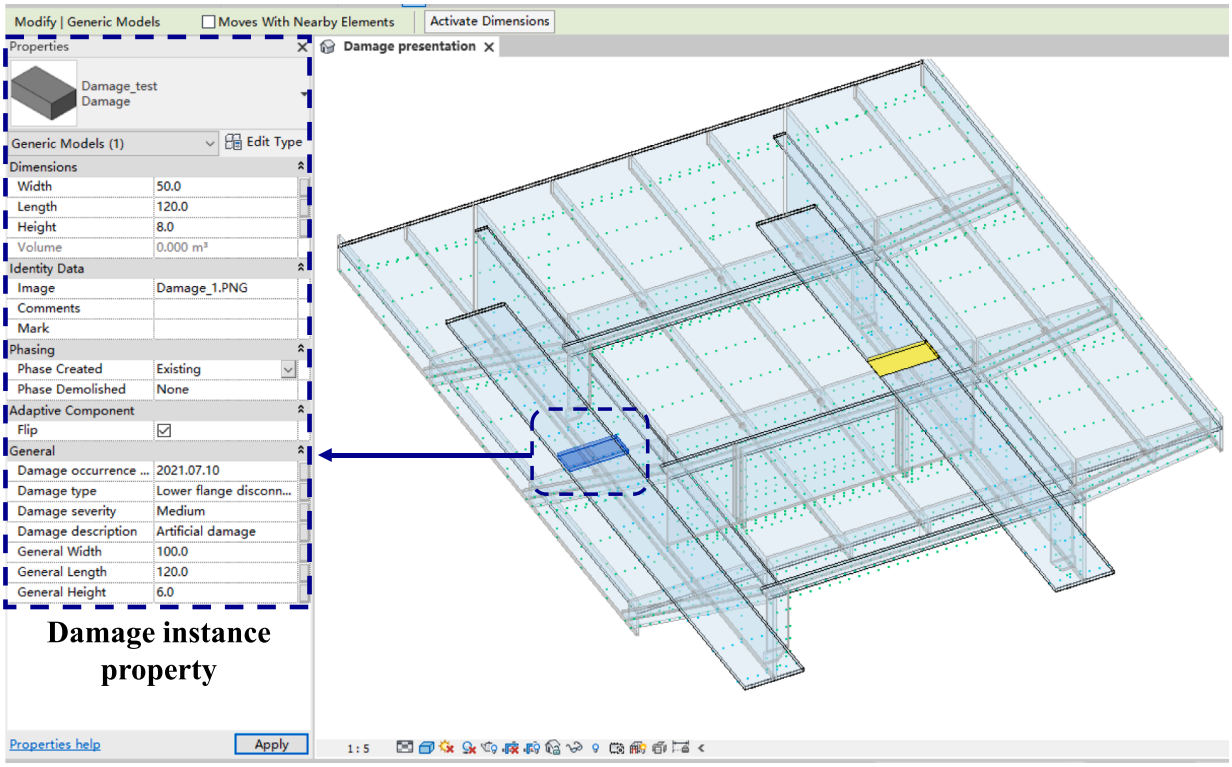


Fig. 18. Damage information recording.

CRediT authorship contribution statement

Limin Sun: Writing – original draft, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization. **Haibin Sun:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Wei Zhang:** Validation, Methodology, Formal analysis, Data curation. **Yixian Li:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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